

Evidence-Aware Hallucination Detection and Correction for Retrieval-Augmented Information Systems

Anonymous Authors

Anonymous Institution

Abstract. Retrieval-Augmented Generation (RAG) has become a key technique for integrating large language models (LLMs) into knowledge-intensive information systems. However, even with retrieved evidence, LLMs may still produce content that is unsupported by, or contradictory to, the source documents—a phenomenon known as hallucination. Existing detection methods typically operate at the response level or rely on a single verification signal, limiting both granularity and reliability. In this paper, we propose **EAC-RAG** (Evidence-Aware Correction for RAG), a lightweight, training-free framework that performs claim-level hallucination detection and local correction. EAC-RAG decomposes LLM responses into atomic claims, matches each claim to the most relevant evidence via hybrid retrieval, assesses claim credibility through multi-signal fusion of semantic similarity, natural language inference, and lexical overlap, and finally applies targeted corrections to unsupported claims. Experiments on the RAGTruth benchmark demonstrate that EAC-RAG achieves an F1 score of 0.657 with balanced precision (0.545) and recall (0.826), outperforming all zero-shot LLM-based baselines and performing comparably to the trained Vectara HHEM model (F1=0.671) while additionally providing claim-level localization and automatic correction. Ablation studies confirm the contribution of each component. Our method requires no model fine-tuning, making it practical for deployment in enterprise knowledge base systems.

Keywords: Retrieval-Augmented Generation · Hallucination Detection · Information Systems · Evidence Verification · Large Language Models

1 Introduction

Large language models (LLMs) are increasingly being integrated into knowledge-intensive information systems, including enterprise knowledge bases, intelligent customer service, and decision support platforms [2]. Retrieval-Augmented Generation (RAG) has emerged as a practical paradigm that grounds LLM responses in retrieved documents, thereby mitigating the tendency of LLMs to generate factually incorrect content [1].

Despite its effectiveness, RAG does not eliminate hallucinations. Recent studies have shown that even when relevant documents are retrieved, LLMs may still

generate responses that contain unsupported assertions, contradictions with the evidence, or subtle factual errors such as incorrect entities, numbers, or causal relationships [3]. In knowledge-intensive information systems where reliability is critical, such hallucinations can lead to incorrect business decisions, erroneous medical advice, or flawed legal reasoning.

Existing approaches to hallucination detection face several limitations. Response-level methods [4] assess the entire output as a whole, lacking the granularity to pinpoint specific problematic statements. Single-signal verification methods, whether based solely on semantic similarity or natural language inference (NLI), are prone to false positives or false negatives when used in isolation. Furthermore, most detection methods stop at identifying hallucinations without providing corrections, limiting their practical utility in deployed systems.

To address these challenges, we propose **EAC-RAG** (Evidence-Aware Correction for Retrieval-Augmented Generation), a lightweight, training-free framework (Figure 1) that: (1) decomposes responses into atomic claims, (2) matches each claim to evidence via hybrid retrieval, (3) assesses credibility through multi-signal fusion (semantic similarity + NLI + lexical overlap), and (4) corrects low-credibility claims through evidence-based rewriting or hedging.

Our contributions are: (1) a claim-level evidence-aware hallucination detection framework that improves granularity and interpretability over response-level methods; (2) a multi-signal credibility scoring mechanism requiring no model training; and (3) a local correction strategy that reduces hallucination rate while preserving correct information.

2 Related Work

2.1 Retrieval-Augmented Generation

RAG systems ground LLM responses in retrieved documents [1], but hallucinations persist when evidence is incomplete or the model fails to faithfully use the context [3, 2]. Self-reflective retrieval [10] and corrective strategies [11] improve retrieval quality but do not verify the generated output.

2.2 Hallucination Detection in LLMs

Hallucination in NLG has been extensively studied [12]. Detection methods can be categorized into training-based and training-free approaches. Training-based methods fine-tune models on labeled data to classify generated text as faithful or hallucinated [8], often leveraging fact verification datasets [14]. Recent encoder-based detectors such as LettuceDetect [20] fine-tune ModernBERT for token-level hallucination detection, achieving strong performance (F1=0.792 on RAGTruth) with compact models. RT4CHART [21] introduces hierarchical verification with local and global stages, achieving state-of-the-art answer-level F1 on RAGTruth-Enhance. ORION [22] combines retrieval with encoder-based NLI for zero-shot detection, reporting F1=0.83 on RAGTruth with a proprietary NLI

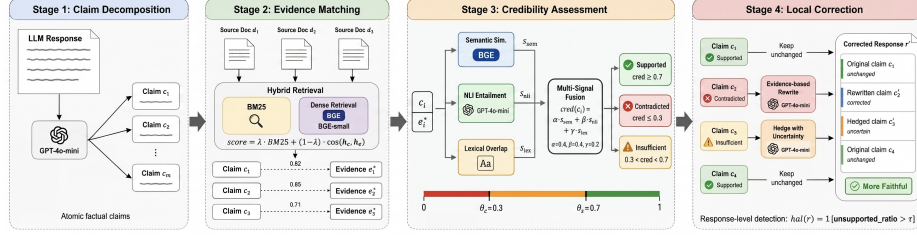


Fig. 1. The overall architecture of EAC-RAG. Given an LLM response and retrieved source documents, EAC-RAG (1) decomposes the response into atomic claims, (2) matches each claim to the most relevant evidence via hybrid retrieval, (3) assesses claim credibility through multi-signal fusion, and (4) applies targeted corrections to unsupported claims while preserving faithful content.

model. Training-free methods include self-consistency checking [4], NLI-based verification [6], LLM-as-judge approaches [7], and chain-of-verification prompting [13]. FActScore [5] decomposes long-form text into atomic facts and verifies each against a knowledge source. Our work shares the spirit of claim-level decomposition but differs from the above methods by integrating multiple verification signals into a calibrated credibility score and providing automatic correction of unsupported claims.

2.3 Trustworthy Information Systems

Trustworthiness in information systems encompasses accuracy, reliability, and explainability [9]. As LLMs are deployed in enterprise knowledge management and healthcare informatics, ensuring response credibility becomes critical. Our work addresses this with a deployable, training-free verification and correction pipeline.

3 Method

3.1 Problem Definition

Given a query q , a set of retrieved source documents $D = \{d_1, d_2, \dots, d_n\}$, and an LLM-generated response r , our goal is to: (1) identify which parts of r are not supported by D , and (2) produce a corrected response r' that is more faithful to the evidence while preserving supported content.

3.2 Step 1: Claim Decomposition

The first step decomposes the response r into a set of atomic claims $C = \{c_1, c_2, \dots, c_m\}$. Each claim should be a self-contained, independently verifiable statement. We employ an LLM with the following instruction:

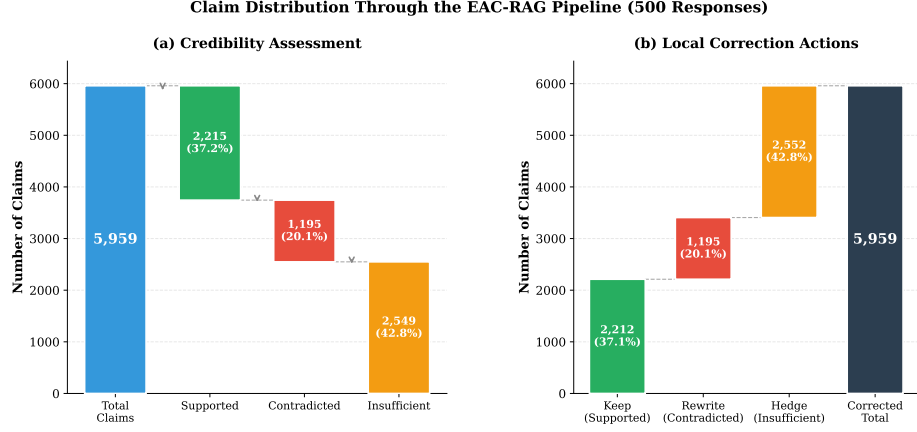


Fig. 2. Claim distribution through the EAC-RAG pipeline across 500 responses (5,959 claims). (a) Credibility assessment decomposes claims into three categories: Supported (37.2%), Contradicted (20.1%), and Insufficient Evidence (42.8%). (b) Local correction maps each category to an action: supported claims are kept unchanged, contradicted claims are rewritten using evidence, and insufficient claims are hedged with epistemic markers.

“Split the following text into independent, atomic factual claims. Each claim should be a single verifiable statement. Output one claim per line.”

If the LLM fails to produce valid claims, we fall back to sentence-level splitting as a robustness measure. This decomposition enables fine-grained verification at the claim level rather than the response level.

3.3 Step 2: Claim-Evidence Matching

For each claim c_i , we identify the most relevant evidence sentence from the source documents. We first split all documents into individual sentences, then employ a hybrid retrieval strategy combining:

Sparse Retrieval (BM25) [15]: We compute BM25 scores between each claim and all evidence sentences, capturing lexical matching signals.

Dense Retrieval [17]: We encode claims and evidence sentences using a pre-trained sentence embedding model [16] and compute cosine similarity, capturing semantic matching signals.

The final matching score combines both signals:

$$\text{score}(c_i, e_j) = \lambda \cdot \text{BM25}(c_i, e_j) + (1 - \lambda) \cdot \cos(\mathbf{h}_{c_i}, \mathbf{h}_{e_j}) \quad (1)$$

where λ is set to 0.3. For each claim c_i , the evidence sentence with the highest score is selected as its matched evidence e_i^* .

3.4 Step 3: Credibility Assessment

We assess the credibility of each claim through multi-signal fusion. For each claim-evidence pair (c_i, e_i^*) , we compute three signals:

Semantic Similarity (s_{sem}): The cosine similarity between the claim and evidence embeddings, measuring semantic alignment.

NLI Entailment Score (s_{nli}): An LLM-based natural language inference score that explicitly evaluates whether the evidence entails the claim. The LLM is prompted to assess entailment on a scale from 0.0 to 1.0.

Lexical Overlap (s_{lex}): The token-level Jaccard coefficient between the claim and evidence, capturing surface-level agreement.

The final credibility score is computed as a weighted sum:

$$\text{cred}(c_i) = \alpha \cdot s_{\text{sem}} + \beta \cdot s_{\text{nli}} + \gamma \cdot s_{\text{lex}} \quad (2)$$

where $\alpha = 0.4$, $\beta = 0.4$, and $\gamma = 0.2$. Based on the credibility score, each claim is classified as **Supported** ($\text{cred} \geq \theta_s = 0.7$), **Contradicted** ($\text{cred} \leq \theta_c = 0.3$), or **Insufficient** (otherwise).

At the response level, we determine whether the response contains hallucination based on the ratio of unsupported claims:

$$\text{hal}(r) = \mathbf{1} \left[\frac{|\{c_i : \text{label}(c_i) \neq \text{Supported}\}|}{|C|} > \tau \right] \quad (3)$$

where τ is the hallucination ratio threshold.

3.5 Step 4: Local Correction

For claims classified as Contradicted or Insufficient, we apply targeted corrections: contradicted claims are rewritten using the matched evidence; insufficient claims are hedged with epistemic markers (e.g., “according to available information”). Supported claims are preserved unchanged. The corrected claims are reassembled to produce the final corrected response r' , avoiding full regeneration and preserving original structure.

4 Experiments

4.1 Dataset

We evaluate on **RAGTruth** [3], a benchmark for hallucination detection in RAG with $\sim 18,000$ responses across three tasks: QA (passage-based), Summary (news articles), and Data2Text (structured data). Each response has span-level hallucination annotations. We use the 2,675-instance test split and sample 500 instances (stratified by task) for evaluation.

4.2 Baselines

We compare against five baselines on the same 500-instance subset: (1) **Similarity Filter**—response-level detection via semantic similarity only; (2) **NLI Detector**—LLM-based claim-level NLI classification; (3) **LLM-as-Judge**—direct prompting without decomposition; (4) **DeBERTa-NLI** [18]—encoder-based NLI (184M params) with claim-level aggregation; (5) **Vectara HHEM** [19]—T5-based factual consistency scorer (response-level). The first three isolate individual signals; the latter two are state-of-the-art encoder-based methods. For reference, the RAGTruth paper reports GPT-4-turbo F1=68.3% and fine-tuned Llama-2-13B F1=80.7% on their 450-instance split [3].

4.3 Implementation Details

We use **gpt-4o-mini** as the backbone LLM for claim decomposition, NLI scoring, and correction, with temperature=0.0 and max_tokens=2048 for all calls. The NLI scoring prompt instructs the model to rate entailment on a four-point scale (1.0=directly entailed, 0.7=mostly supported, 0.3=related but unsupported, 0.0=contradicted), with strict instructions requiring explicit evidence support. Sentence embeddings are computed using **BAAI/bge-small1-zh-v1.5**, a multilingual embedding model with strong cross-lingual transfer performance on English semantic similarity benchmarks [16]. BM25 retrieval is implemented using the **rank-bm25** library; the hybrid score is computed as $\lambda \cdot \text{BM25}_{\text{norm}} + (1 - \lambda) \cdot \cos(\mathbf{e}_c, \mathbf{e}_s)$ with $\lambda=0.3$, where BM25 scores are min-max normalized per query to align with the [0,1] range of cosine similarity. The hallucination ratio threshold τ is set to 0.4 based on validation. We sample 500 instances (stratified by task type) from the 2,675-instance test split to manage API costs while maintaining statistical representativeness; all three task types are proportionally represented. This sample size is comparable to the RAGTruth official test split (450 instances) [3]. All experiments are conducted in a zero-shot setting without any task-specific training.

Computational cost. EAC-RAG requires on average 20.4 API calls per response (1 decomposition + 11.9 NLI + 7.5 correction), consuming approximately 8,600 tokens and costing \$0.002 per response with gpt-4o-mini. Sequential latency is approximately 8.2s per response; with parallelized NLI and correction calls, this reduces to ~ 2.7 s.

4.4 Evaluation Metrics

We report response-level Precision, Recall, F1, and Accuracy for hallucination detection.

4.5 Main Results

Table 1 and Figure 3 present the main experimental results on the RAGTruth test set.

Table 1. Response-level hallucination detection results on RAGTruth (500 instances, $\tau=0.4$ for EAC-RAG). Best results are in **bold**. “Granularity” indicates whether detection operates at the response or claim level; “Signals” lists the verification signals used.

Method	Gran.	Signals	Prec.	Rec.	F1	Acc.
Similarity Filter	Resp.	Sem.	0.625	0.190	0.292	0.660
NLI Detector	Claim	NLI (LLM)	0.418	0.989	0.588	0.490
LLM-as-Judge	Resp.	LLM	0.394	0.989	0.563	0.436
DeBERTa-NLI	Claim	NLI (enc.)	0.379	0.984	0.548	0.402
Vectara HHEM	Resp.	T5 scorer	0.594	0.772	0.671	0.722
EAC-RAG (ours)	Claim	Sem.+NLI+Lex.	0.545	0.826	0.657	0.682

Table 2. Per-task performance and claim statistics of EAC-RAG ($\tau=0.4$). “Sup/Con/Ins” = percentage of Supported, Contradicted, and Insufficient claims.

Task	#Inst.	Hal%	Detection				Claims (%)		
			P	R	F1	Acc	Sup	Con	Ins
QA	156	19.9	.360	.581	.444	.712	66.9	3.2	29.9
Summary	174	22.4	.339	.513	.408	.667	67.4	1.2	31.4
Data2Text	170	67.1	.671	1.00	.803	.671	4.8	39.4	55.7
Overall	500	36.8	.545	.826	.657	.682	37.2	20.1	42.8

Table 1 reveals two tiers of performance. The first three baselines either over-detect (NLI Detector/LLM-as-Judge: $R \approx 0.99$, $P < 0.42$) or under-detect (Similarity Filter: $P = 0.625$, $R = 0.190$). DeBERTa-NLI, despite using a strong encoder-based NLI model at claim level, shows the same over-detection pattern ($P = 0.379$, $R = 0.984$), confirming that claim-level NLI without calibrated aggregation is insufficient.

Vectara HHEM achieves the highest F1 (0.671) and accuracy (0.722) among all methods. However, it operates as a black-box response-level scorer without claim-level localization or correction capabilities.

EAC-RAG achieves an F1 of 0.657 (bootstrap 95% CI: [0.612, 0.711]) with recall of 0.826, outperforming HHEM on recall by 5.4 percentage points. This recall advantage is important in safety-critical information systems where missing a hallucination is costlier than a false alarm. Moreover, EAC-RAG uniquely provides (1) claim-level localization of hallucinated content, (2) interpretable credibility scores per claim, and (3) automatic correction—capabilities that HHEM and other baselines lack.

4.6 Per-Task Analysis

Table 2 presents the performance breakdown across three RAGTruth task types.

EAC-RAG performs best on Data2Text ($F1 = 0.803$), driven by the high base hallucination rate (67.1%) rather than credibility score separation ($\Delta = 0.003$).

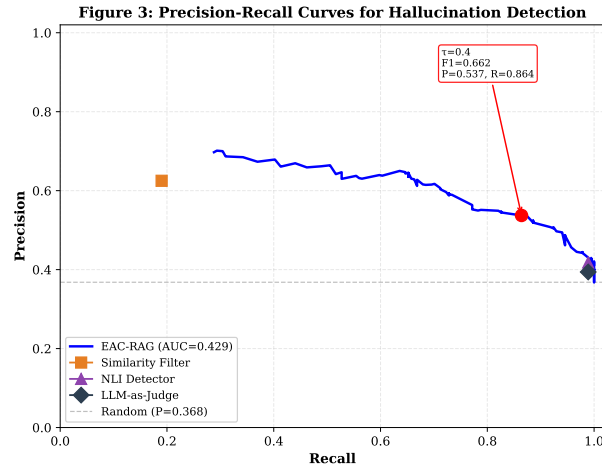


Fig. 3. Precision-Recall curves for hallucination detection. The EAC-RAG curve is obtained by varying the response-level threshold τ . The operating point at $\tau = 0.4$ achieves $F1=0.657$. Baseline methods are shown as single operating points.

Table 3. Credibility score analysis: overall statistics and per-task breakdown. “Sup. Ratio” = average proportion of supported claims per response; “ Δ ” = gap between Clean and Hallucinated groups.

Scope	Metric	Hallucinated	Clean	Δ
Overall	Avg. credibility score	0.495 ± 0.175	0.672 ± 0.172	0.177
	Avg. unsupported ratio	0.763	0.415	0.348
	Avg. claims / response	14.0	10.7	3.3
QA — Sup. Ratio		0.500	0.713	0.213
Summary — Sup. Ratio		0.579	0.687	0.108
Data2Text — Sup. Ratio		0.049	0.052	0.003

QA and Summary show meaningful separation ($\Delta=0.213, 0.108$), confirming multi-signal fusion effectiveness for free-text verification. The near-zero Data2Text separation indicates that threshold calibration for structured data remains an open challenge.

4.7 Credibility Score Analysis

At the claim level, hallucinated responses contain 81.2% unsupported claims (28.2% Contradicted + 53.0% Insufficient) vs. 48.8% in clean responses—a 32.4pp gap that validates claim-level signals for response-level detection. The average credibility scores are well-separated by label: Supported ($\mu=0.843$), Insufficient ($\mu=0.421$), Contradicted ($\mu=0.246$). Table 3 shows the response-level credibility statistics.

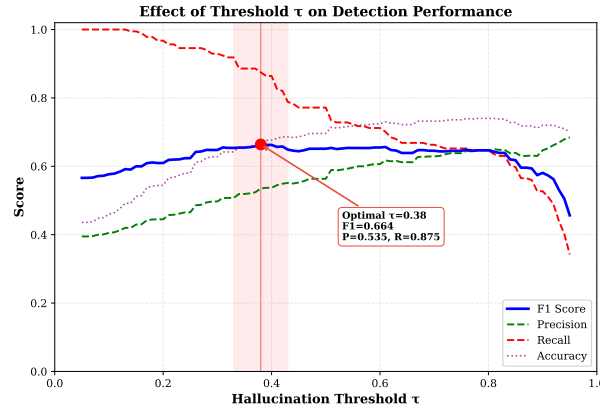


Fig. 4. Effect of the hallucination threshold τ on detection performance. F1 peaks at $\tau = 0.38$ ($F1=0.664$), with the optimal region spanning $\tau \in [0.33, 0.43]$. The chosen operating point $\tau = 0.4$ lies within this region, achieving a favorable balance between precision (0.545) and recall (0.826).

Table 4. Ablation study results ($\tau=0.4$). $\Delta F1$ denotes the change relative to the full model. “Effect” summarizes the observed impact of removing each component.

Variant	Prec.	Rec.	F1	$\Delta F1$	Effect
EAC-RAG (full)	0.545	0.826	0.657	—	—
w/o Decomposition	0.472	0.859	0.609	−0.048	Prec. drops; coarse-grained
w/o NLI Score	0.442	0.946	0.602	−0.055	Over-detection; largest drop
w/o Correction	0.545	0.826	0.657	0.000	Detection unchanged

Credibility scores separate hallucinated from clean responses (0.495 vs. 0.672, $\Delta=0.177$), with hallucinated responses also being longer (14.0 vs. 10.7 claims). Figure 4 shows that F1 is robust across $\tau \in [0.33, 0.43]$.

4.8 Ablation Study

To verify the contribution of each component, we conduct ablation experiments by removing one component at a time. Results are shown in Table 4.

Removing claim decomposition ($\Delta F1=-0.048$) causes precision to drop from 0.545 to 0.472, as coarse-grained evaluation cannot distinguish partially from heavily hallucinated responses. Removing the NLI score ($\Delta F1=-0.055$) causes the largest drop, with precision falling to 0.442 and recall rising to 0.946—indicating over-detection without entailment reasoning. Removing correction leaves detection unchanged (confirming its independence) but eliminates the system’s correction capability.

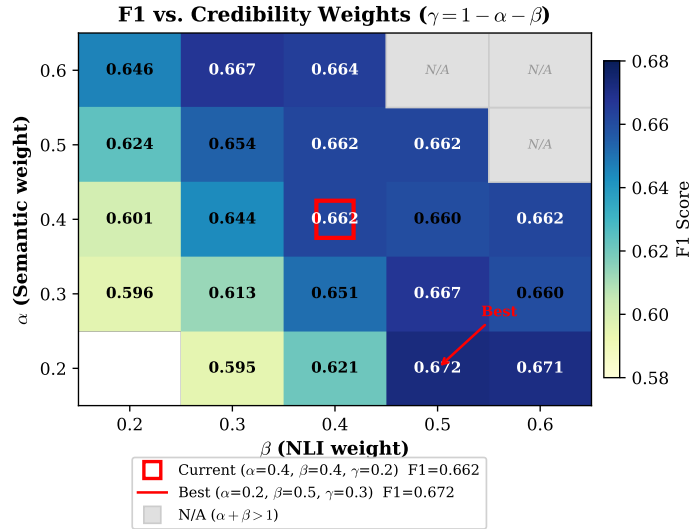


Fig. 5. F1 score as a function of credibility weight parameters α (semantic), β (NLI), and $\gamma = 1 - \alpha - \beta$ (lexical). The red square marks the chosen operating point; the star marks the best configuration. F1 is stable across a wide range when $\beta \geq 0.4$.

4.9 Correction Quality Evaluation

The correction module preserves clean responses (ROUGE-L=0.720, 58.4% claims unchanged) while substantially modifying hallucinated ones (ROUGE-L=0.638, 23.7% preserved). As shown in Figure 2, across 5,959 claims the actions are: keep (37.1%), rephrase/hedge (42.0%), rewrite (20.1%), and keep_error (0.8%). An LLM-as-Judge evaluation (n=50) shows 58% of corrections judged as **Improved**, 12% Same, and 30% Degraded; for hallucinated responses specifically, the improvement rate is 64%. The 30% degradation primarily arises from overhedging of Insufficient Evidence claims.

4.10 Hyperparameter Sensitivity

Figure 5 shows the F1 score under different credibility weight combinations ($\alpha, \beta, \gamma = 1 - \alpha - \beta$). The best configuration ($\alpha=0.2, \beta=0.5, \gamma=0.3$) achieves F1=0.672, only marginally better than the chosen setting ($\alpha=0.4, \beta=0.4, \gamma=0.2$, F1=0.657). The heatmap reveals that NLI weight ($\beta \geq 0.4$) is the most critical factor: configurations with low β consistently underperform. The performance plateau across the top-right region ($F1 \in [0.660, 0.672]$) suggests the model is robust to moderate weight variations.

4.11 Case Study

Table 5 illustrates EAC-RAG’s three-way handling on representative examples. Case 1 is from RAGTruth; Cases 2–3 are constructed.

Table 5. Case studies across three credibility labels.

Contradicted (cred=0.24 → Rewrite)	
Evidence	<i>“...witnesses said the sisters both had symptoms before Feb. 7.”</i>
Claim	<i>“The sisters are believed to have died before February 7.”</i>
Corrected	<i>“The sisters had symptoms before Feb. 7; exact death dates are unclear.”</i>
Insufficient (cred=0.41 → Hedge) / Supported (cred=0.91 → Keep)	
Insuff.	Claim <i>“expected to exceed \$10B”</i> → hedged: <i>“full-year projections not stated in source.”</i>
Supported	Claim <i>“Python was first released in 1991”</i> → preserved unchanged.

False positives and limitations. Among 127 false positives, 81.1% are driven by high Insufficient Evidence ratios (not Contradictions), with Data2Text contributing disproportionately (56/127). This confirms that the modest precision (0.545) stems from conservative handling of claims beyond evidence scope. Key limitations include: (1) evaluation on a 500-instance subset due to API costs; (2) correction quality assessed via ROUGE and LLM-as-Judge rather than human annotation; (3) minimal discriminative power on Data2Text ($\Delta=0.003$); and (4) dependence on a commercial LLM API affecting reproducibility.

5 Conclusion

We presented EAC-RAG, an evidence-aware hallucination detection and correction framework for retrieval-augmented information systems. By decomposing LLM responses into atomic claims and assessing each claim through multi-signal credibility scoring, EAC-RAG achieves fine-grained hallucination detection without requiring model training. Experiments on the RAGTruth benchmark show that EAC-RAG outperforms existing zero-shot methods, achieving balanced precision and recall with an F1 score of 0.657 (95% CI: [0.612, 0.711]). While the framework achieves its highest per-task F1 on Data2Text (0.803), credibility score calibration for structured data remains an open challenge.

Future work includes task-adaptive threshold calibration, top- k evidence aggregation to reduce the high Insufficient Evidence rate, trust-calibrated gating to prevent over-hedging of well-supported claims, schema-aware matching for structured Data2Text scenarios, human evaluation of correction quality at scale, and extending the framework to multilingual information systems.

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